LBNL IC GROUP DESIGN NOTE

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LArPix-v2b Datasheet

The Liquid Argon Pixel readout ASIC, Version 2b (LArPix-v2b) is a data acquisition ASIC intended for use with a 2D liquid argon time-project chamber (LAr-TPC). 2D readout is a proposed technology for the DUNE Near Detectors. The datasheet is intended to communicate the functionality and use of LArPix-v2b. LArPix-v2b is implemented in 180 nm CMOS technology.

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REVISION NUMBER	KEY CHANGES	ENGINEER
1.0	Initial draft.	CRG
1.1	Clarified configuration of new digital I/O.	CRG

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1 Document Description

The Datasheet for Version 2b of the Liquid Argon Pixel (LArPix-v2b) readout ASIC is intended to communicate the functionality and use of LArPix-v2b. Details relevant for applying LArPix-v2b in a system are highlighted.

While LArPix-v2b is a functional upgrade from LArPix-v2, this document will include all relevant detail to operate LArPix-v2b. In other words, there is a lot of overlapping information between the LArPix-v2 and LArPix-v2b datasheets, but the LArPix-v2 datasheet will not be required to operate LArPix-v2b.

2 LArPix-v2b

2.1 Differences between LArPix-v2 and LArPix-v2b

There are several differences between LArPix-v2 and LArPix-v2b. Beyond bug fixes, the key change is that the digital I/O has been converted from single ended to differential. This includes not only the digital communication links (PISO/POSI) but also the input clocks and fast command interfaces (clock,

reset, external trigger). This change was made primarily to address issues with crosstalk that were caused by the use of single-ended signaling. From the system designer's standpoint, the main differences are the need to update any system designs to accommodate the differential paths and the updated padring. From the user's standpoint, the main differences are the updated configuration map and the need to configure the new digital transceivers.

2.2 LArPix-v2b Concept

The LAr Pixel ASIC, Version 2 (LArPix-v2b) is a data acquisition ASIC intended for use with pixelated 2-D liquid argon time-project chambers (LAr-TPCs). Two-dimensional readout is a proposed technology for the DUNE Near Detectors. The basic idea is to implement a 2D area detector using a PCB with vias as the pixel sensors. Biased strips on the PCB focus electrons onto the vias and the collected charge is read out using LArPix-v2b.

LArPix-v2b contains 64 analog channels. LArPix-v2b is organized conceptually in quadrants, and the individual channels are distributed around the chip as shown in Figure 1. There are 16 channels on each side as well as a digital I/O block (labeled POSI/PISO) in each corner of the chip. Each channel comprises a charge-sensitive amplifier (CSA), a discriminator with a programmable threshold, and an 8-bit analog-to-digital converter (ADC). Charge incident on the input of each channel is converted by the CSA to a voltage that is proportional to the charge received since the last reset pulse. If this voltage is larger than the threshold, the discriminator fires and an ADC conversion is initiated. The ADC output is then assembled into an event by the digital core. In addition to the ADC value, each event includes the channel and chip IDs as well as a time stamp.

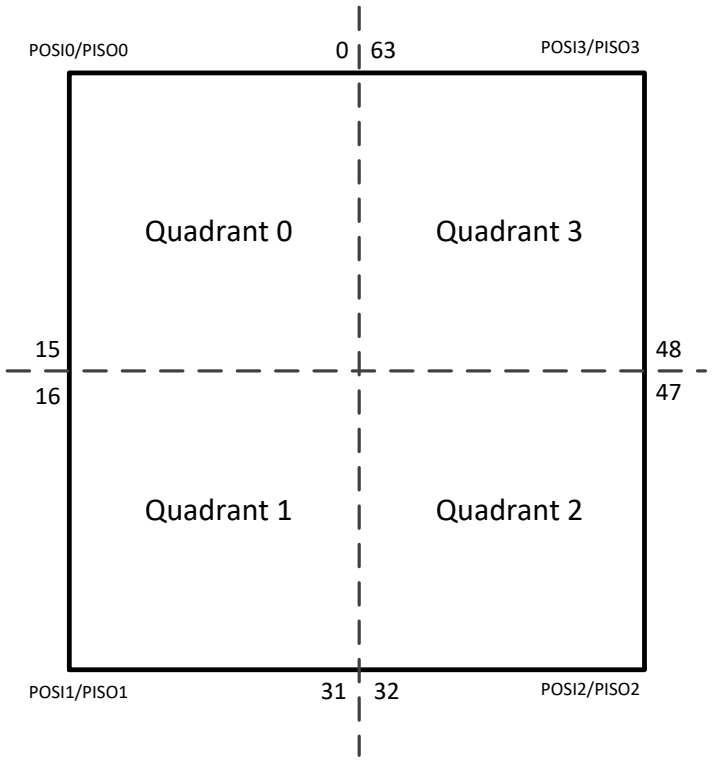


Figure 1. Quadrant Definitions for LArPix-v2b.

A block diagram of LArPix_v2b channel allocation is shown in Figure 1. The 64 input channels are organized in four independent banks of 16 channels each. The POSI/PISO connections can be defined as desired by the system designer (as long as the designer is consistent). As stated above, please note that the POSI/PISO quadrants have independent power connections and must all be connected externally.

To reduce the number of connections external to the cryostat, a collection of LArPix-v2b chips are organized as a network with a single LArPix-v2b chip functioning as the interface between the network and the control electronics. This functionality is called “Hydra-I/O” and is described in more detail in its own section.

In addition to data-driven operation, LArPix-v2b also includes programmable external trigger and a cross-trigger function, where all channels will convert if a hit occurs in any channel.

3 LArPix-v2b Pads (Bare Die)

The LArPix-v2b padring is shown in Figure 2. LArPix-v2b has 204 pads arrayed on a 92 µm pitch. Each pad is 95.24 µm long and 50 µm wide. The pads for the bare die are numbered counter-clockwise with pad #1 the top pad on the left side of the chip. The pads are listed in Table 3.

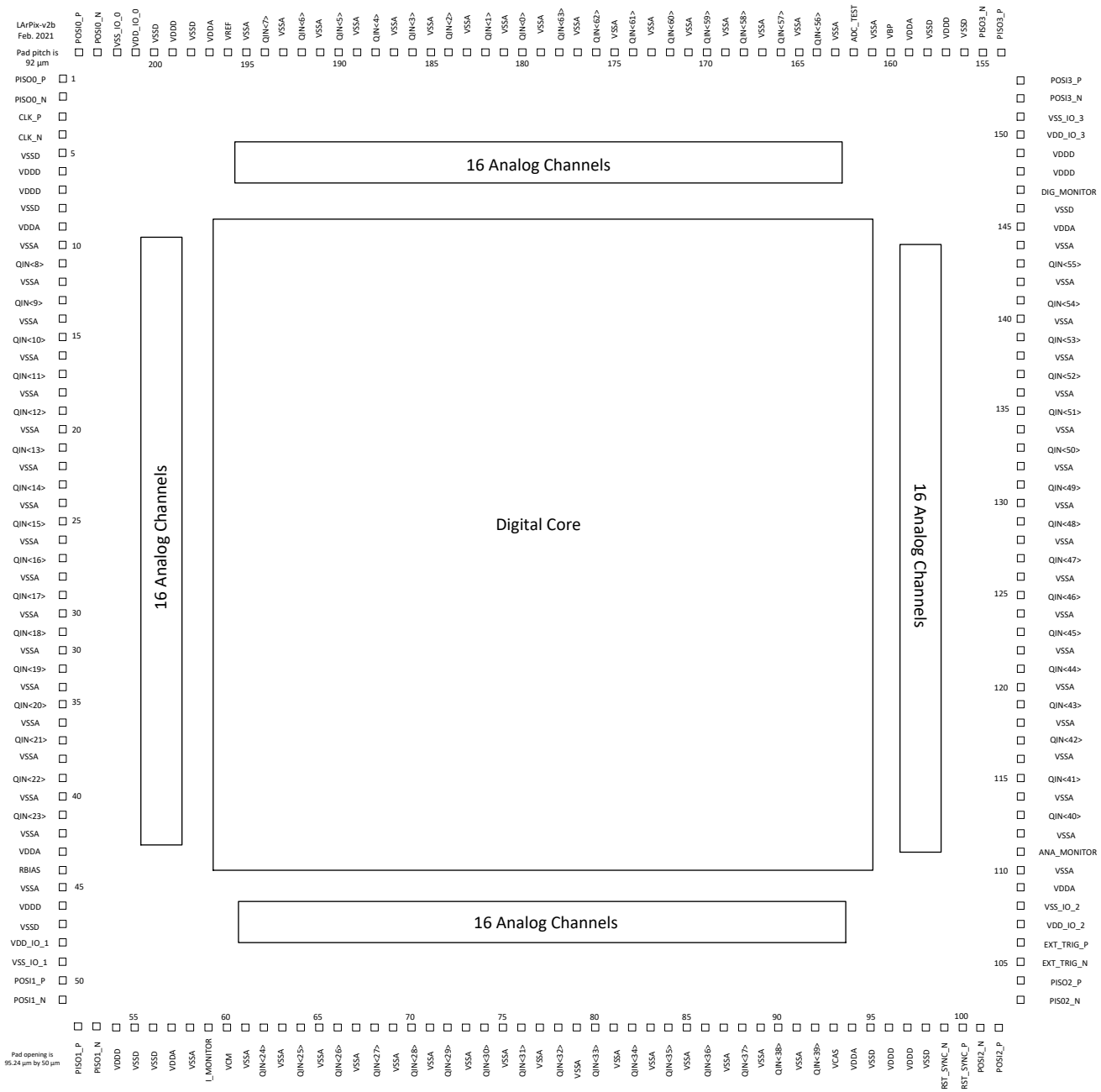


Figure 2. LArPix-v2b padring.

PAD	PAD NAME	PAD	PAD NAME	PAD	PAD NAME	PAD	PAD NAME
1	PISO0_P	52	PISO1_P	103	PISO2_N	154	PISO3_P
2	PISO0_N	53	PISO1_N	104	PISO2_P	155	PISO3_N
3	CLK_P	54	VDDD	105	EXT_TRIG_N	156	VSSD
4	CLK_N	55	VSSD	106	EXT_TRIG_P	157	VDDD
5	VSSD	56	VSSD	107	VDD_IO_2	158	VSSD
6	VDDD	57	VDDA	108	VSS_IO_2	159	VDDA
7	VDDD	58	VSSA	109	VDDA	160	VBP
8	VSSA	59	I_MONITOR	110	VSSA	161	VSSA
9	QIN<8>	60	VCM	111	ANA_MONITOR	162	ADC_TEST
10	VSSA	61	VSSA	112	VSSA	163	VSSA
11	QIN<8>	62	QIN<24>	113	QIN<40>	164	QIN<56>
12	VSSA	63	VSSA	114	VSSA	165	VSSA
13	QIN<9>	64	QIN<25>	115	QIN<41>	166	QIN<57>
14	VSSA	65	VSSA	116	VSSA	167	VSSA
15	QIN<10>	66	QIN<26>	117	QIN<42>	168	QIN<58>
16	VSSA	67	VSSA	118	VSSA	169	VSSA
17	QIN<11>	68	QIN<27>	119	QIN<43>	170	QIN<59>
18	VSSA	69	VSSA	120	VSSA	171	VSSA
19	QIN<12>	70	QIN<28>	121	QIN<44>	172	QIN<60>
20	VSSA	71	VSSA	122	VSSA	173	VSSA
21	QIN<13>	72	QIN<29>	123	QIN<45>	174	QIN<61>
22	VSSA	73	VSSA	124	VSSA	175	VSSA
23	QIN<14>	74	QIN<30>	125	QIN<46>	176	QIN<62>
24	VSSA	75	VSSA	126	VSSA	177	VSSA

25	QIN<15>	76	QIN<31>	127	QIN<47>	178	QIN<63>
26	VSSA	77	VSSA	128	VSSA	179	VSSA
27	QIN<16>	78	QIN<32>	129	QIN<48>	180	QIN<0>
28	VSSA	79	VSSA	130	VSSA	181	VSSA
29	QIN<17>	80	QIN<33>	131	QIN<49>	182	QIN<1>
30	VSSA	81	VSSA	132	VSSA	183	VSSA
31	QIN<18>	82	QIN<34>	133	QIN<50>	184	QIN<2>
32	VSSA	83	VSSA	134	VSSA	185	VSSA
33	QIN<19>	84	QIN<35>	135	QIN<51>	186	QIN<3>
34	VSSA	85	VSSA	136	VSSA	187	VSSA
35	QIN<20>	86	QIN<36>	137	QIN<52>	188	QIN<4>
36	VSSA	87	VSSA	138	VSSA	189	VSSA
37	QIN<21>	88	QIN<37>	139	QIN<53>	190	QIN<5>
38	VSSA	89	VSSA	140	VSSA	191	VSSA
39	QIN<22>	90	QIN<38>	141	QIN<54>	192	QIN<6>
40	VSSA	91	VSSA	142	VSSA	193	VSSA
41	QIN<23>	92	QIN<39>	143	QIN<55>	194	QIN<7>
42	VSSA	93	VCAS	144	VSSA	195	VSSA
43	VDDA	94	VDDA	145	VDDA	196	VREF
44	RBIAS	95	VSSD	146	VSSD	197	VDDA
45	VSSA	96	VDDD	147	DIG_MONITOR	198	VSSD
46	VDDD	97	VDDD	148	VDDD	199	VDDD
47	VSSD	98	VSSD	149	VDDD	200	VSSD
48	VDD_IO_1	99	RST_SYNC_N	150	VDD_IO_3	201	VDD_IO_0
49	VSS_IO_1	100	RST_SYNC_P	151	VSS_IO_3	202	VSS_IO_0

50	POSI1_P	101	POSI2_N	152	POSI3_N	203	POSI0_N
51	POSI1_N	102	POSI2_P	153	POSI3_P	204	POSI0_P

Table 2. LArPix-v2b padring.

The function of each pad is described in Table 3, including information on required pad connections. The digital I/O circuits used the VDD_IO_X and VSS_IO_X connections. Please note the difference VDD_IO_X domains (e.g. VDD_IO_0, VDD_IO_1, etc) are not shorted together internally. Therefore, each pad connection must be made for LArPix-v2b to function correctly. The internal digital circuits use the nominally 1.8 V VDDD supply which can be lowered to reduce power dissipation. VDDA can be also be lowered from its nominal level of 1.8 V to reduce power dissipation at the expense of dynamic range.

Pad Name	**Direction**	**Type**	**Description**	**Number of Pads**	**Bypass**
QIN[63:0]	IN	Analog	Charge inputs from sensor	64	
VDDA	INOUT	Power	1.8 V (nominal) Analog Power for Channel	9	0.1 µF in parallel with 1 µF to VSS
VDDD	INOUT	Power	1.8 V (nominal) Digital Power	8	0.1 µF in parallel with 1 µF to VSS
VSSA	INOUT	Power	Analog Ground	70	
VSSD	INOUT	Power	Digital Ground	8	
PISO0[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 0. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	
PISO1[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 1. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	

PISO2[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 2. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	
PISO3[3:0]_{P/N}	OUT	Digital	Primary-In-Secondary-Out, port 3. Differential current-mode digital data output from current chip to other chips or FPGA in Hydra network	2	
POSI0[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 0. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
POSI1[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 1. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
POSI2[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 2. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
POSI3[3:0]_{P/N}	IN	Digital	Primary-Out-Secondary-In, port 3. Differential, current-mode digital input from other chips or FPGA in Hydra network to current chip. Internally terminated.	2	
CLK_{P/N}	IN	Digital	Differential, current mode	2	

			digital clock input. Can be internally or externally terminated.		
EXTERN_TRIG_{P/N}	IN	Digital	Differential, current-mode external trigger to initiate conversion (active high). Can be internally or externally terminated.	2	
RST_SYNC_{P/N}	IN	Digital	Differential, current-mode synchronous global reset (active low). Also provides configuration reset (soft reset) and timestamp synchronization. Can be internally or externally terminated.	1	
ANA_MONITOR	OUT	Analog	Monitor port for various internal analog signals.	1	
DIG_MONITOR	OUT	Digital	Monitor port for various internal digital signals.	1	
I_MONITOR	OUT	Analog	Monitor port for various internal bias currents. Must be terminated with appropriate resistor to ground (monitor bus is a current source).	1	
VREF	INOUT	Analog	External bypass connection pin for VREF.	2	0.1 µF in parallel with 10 µF to VSS
VCM	INOUT	Analog	External bypass connection pin for VCM.	3	0.1 µF in parallel with 10 µF to VSS
RBIAS	INOUT	Analog	Optional bias resistor connection for LArPix-v2b biasing.	1	
VCAS	INOUT	Analog	Internal node brought out for optional filtering. Nominally a	1	

			no-connect.		
VBP	INOUT	Analog	Internal node brought out for optional filtering. Nominally a no-connect.	1	
ADC_TEST	IN	Analog	Test voltage signal for ADC	1	

Table 3. List of LArPix-v2b pads.

4 LArPix-v2b Package (OBSOLETE – TBD)

LArPix-v2b is also available in a TBD package. A schematic of the package showing which pads are bonded out to package pins is shown in Figure 3.

THIS SECTION IS CURRENTLY OBSOLETE. TO BE COMPLETED WHEN PACKAGING INFORMATION IS AVAILABLE.

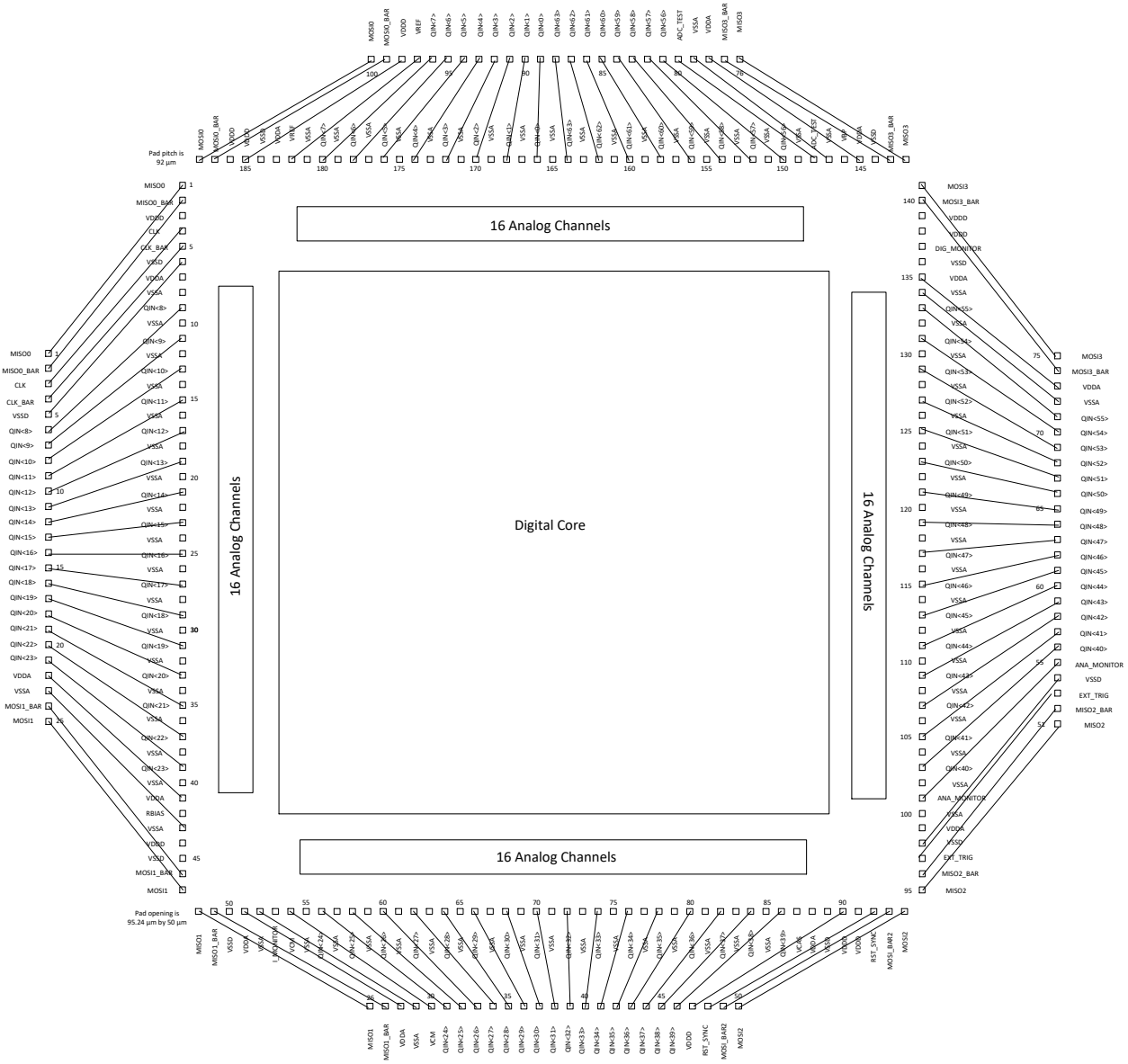


Figure 3. LArPix-v2b QFP100 Package.

The QFP100 pins are listed in Table 4 starting in the upper left-hand corner and proceeding counter-clockwise. The functions and suggested bypassing are the same as for the bare die version of LArPix-v2b.

PIN	PIN NAME	PIN	PIN NAME	PIN	PIN NAME	PIN	PIN NAME
1	PISO0	26	PISO1	51	PISO2	76	PISO3

2	PISO0_BAR	27	PISO1_BAR	52	PISO2_BAR	77	PISO3_BAR
3	CLK	28	VDDA	53	EXT_TRIG	78	VDDA
4	CLK_BAR	29	VSSA	54	VSSD	79	VSSA
5	VSSD	30	VCM	55	ANA_MONITOR	80	ADC_TEST
6	QIN<8>	31	QIN<24>	56	QIN<40>	81	QIN<56>
7	QIN<9>	32	QIN<25>	57	QIN<41>	82	QIN<57>
8	QIN<10>	33	QIN<26>	58	QIN<42>	83	QIN<58>
9	QIN<11>	34	QIN<27>	59	QIN<43>	84	QIN<59>
10	QIN<12>	35	QIN<28>	60	QIN<44>	85	QIN<60>
11	QIN<13>	36	QIN<29>	61	QIN<45>	86	QIN<61>
12	QIN<14>	37	QIN<30>	62	QIN<46>	87	QIN<62>
13	QIN<15>	38	QIN<31>	63	QIN<47>	88	QIN<63>
14	QIN<16>	39	QIN<32>	64	QIN<48>	89	QIN<0>
15	QIN<17>	40	QIN<33>	65	QIN<49>	90	QIN<1>
16	QIN<18>	41	QIN<34>	66	QIN<50>	91	QIN<2>
17	QIN<19>	42	QIN<35>	67	QIN<51>	92	QIN<3>
18	QIN<20>	43	QIN<36>	68	QIN<52>	93	QIN<4>
19	QIN<21>	44	QIN<37>	69	QIN<53>	94	QIN<5>
20	QIN<22>	45	QIN<38>	70	QIN<54>	95	QIN<6>
21	QIN<23>	46	QIN<39>	71	QIN<55>	96	QIN<7>
22	VDDA	47	VDDD	72	VSSA	97	VREF
23	VSSA	48	RST_SYNC	73	VDDA	98	VDDD
24	POSI1_BAR	49	POSI2_BAR	74	POSI3_BAR	99	POSI0_BAR

25	POSI1	50	POSI2	75	POSI3	100	POSI0

Table 4. List of LArPix-v2b QFP100 pins.

5 Digital Electrical Interface

5.1 Clock Definitions

LArPix-v2b receives a differential CMOS system clock called CLK. Data is transmitted on- and off-chip using a divided down version of the system clock. The division ratio is nominally 2 but can be adjusted via the configuration interface. This divided down clock is called the transmit clock.

5.2 UART

LArPix-v2b uses a simple, industry-standard serial protocol to transfer data on- and off-chip. To implement this protocol at low power, LArPix-v2b contains a Universal Asynchronous Receiver/Transmitter (UART). LArPix-v2b sends and receives data in a bit-serial fashion. The input port of the UART is called the Primary-Out-Secondary-In (POSI), and the transmit port is called the Primary-In-Secondary-Out (PISO). In order to coordinate the data transmission, the UART uses a self-synchronizing start- and stop-bit data format as shown in Figure 4.



Figure 4. UART data format.

Each UART word comprises one leading low start bit, one trailing high stop bit, and 64 data bits. The UART does not implement a parity bit directly but parity is implemented in the event building process. The UART operates as follows. When no data transmission is taking place, the output line of the UART is held high. The start bit of a data transmission sequence is indicated by the output falling low. Then the data is sent out one bit per baud[1]. The UART sends bit [0] of the data word first and bit [53] is sent last.

[1] The baud rate in LArPix-v2b is equal to the transmit clock rate (nominally the system clock divided by two).

After all 64 data bits are sent, the stop bit is indicated by the output going high. The output is then kept high until the next start bit indicates transmission of a new data word. The contents of the UART word are described in the section on the LArPix-v2b configuration interface.

The UART input to LArPix-v2b is oversampled by the system clock and each data bit should be launched by the falling edge of the system clock, as shown in Figure 5.

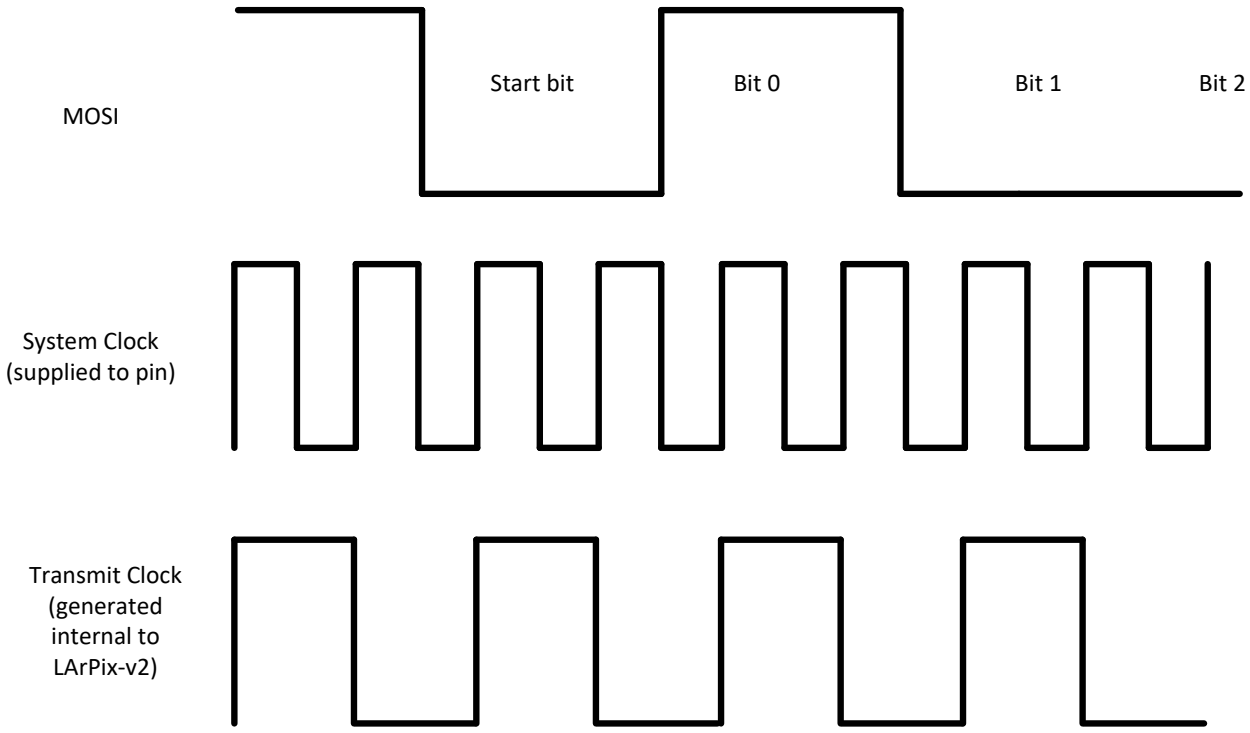


Figure 5. LArPix-v2b UART timing diagram.

5.3 Reset/Sync Timing

LArPix-v2b can execute three operations through assertion of the active-low RST_SYNC pad (in other words, it is kept high naturally and pulled low to assert it). The behavior of RST_SYNC is summarized in Table 5.

Number of clock cycles RST_SYNC held low	**Action**

More than 4	Sync timestamp (set to 0)
More than 15	Internal reset (reset state machines but do not reset configuration memory)
More than 32	Full reset (reset all internal state machines as well as reset configuration memory)

Table 5. RST_SYNC behavior.

6 LArPix-v2b Specifications

6.1 Power Supplies

The power supply specifications for LArPix-v2b are summarized in Table 6.

Specification	**Padring Name**	**Value**	**Units**	**Note**
Analog Power Supply	VDDA	1.8 nominal	V	VDDA → 1.9 mA @ 1.8V
Digital Logic Power Supply	VDDD	1.8 nominal (functional down to 1 V)	V	VDDD → 5 mA @ 10 MHz System Clock, 1.8 V supply
Total Power Dissipation		~ 3.4 mW analog ~ 9 mW digital ---------- 12.4 mW total (~ 200 µW/channel)		150 µW / channel @ 300 K & 10 MHz CLK operation (1 Hz average hit rate) Can be significantly improved by reducing operating voltages.

Table 6. LArPix-v2b power supply specifications.

6.2 Analog Specifications

The analog specifications for LArPix-v2b are summarized in Table 7.

Specification	Value	Units	Note
Number of Analog Inputs (channels)	64 (single-ended)		
Noise	300 @ 88K 500 @ 300K	ENC	
Channel gain	4 or 2	µV/e-	Digitally programmable
Time resolution	2	µs	with 10 MHz clock rate
Analog Dynamic Range	~1300	mV	max signal ~ 20 MIP, minimum detectable signal ~0.04 MIP (+ 6 dB rms noise)
ADC resolution	8	bits	programmable LSB, 4 mV nominal (= 0.1 fC or 600 e-)
Threshold Range	0 – 1.8	V	
Threshold Resolution	< 1	mV	nominal
Channel Linearity	1	%	
Operating Temperature Range	80 - 300	°K	

Table 7. Analog Specifications.

6.3 Digital Specifications

The digital specifications for LArPix-v2b are summarized in Table 8.

Specification	Value	Units	Note
Event Memory Depth	2048	memory locations	~8 ms without data loss in case of track normal to pixel plane
Output Signaling Level	Programmable	V	Programmable, differentially LVDS-like signaling.

Digital data rate	5	Mb/s	with 10 MHz System Clock
Event readout time	12.8	µs	

Table 8. Digital Specifications.

7 LArPix-v2b Configuration

7.1 LArPix-v2b Channel

The high-level design for the LArPix-v2b ASIC is similar to many readout ASICs for capacitive radiation detectors with the exception that a shaping amplifier is not included in LArPix-v2b. This is possible because the noise specifications of the ASIC are significantly relaxed because the detector capacitance is much less than expected for the LArASIC ASIC, for instance. Removing the shaper not only greatly reduces power dissipation it also simplifies and accelerates the design of the ASIC as many important and challenging design considerations are eliminated. A conceptual block diagram of the channel is shown in Figure 6.

The channel operates as follows. The charge input from the detector is integrated on the feedback capacitance (labeled C) of a charge-sensitive amplifier (CSA). The CSA followed by a buffer amplifier and the buffer output is fed to the input of a discriminator with a per-channel programmable threshold and to the ADC. The discriminator fires if the CSA output reaches the point where it exceeds the discriminator threshold. The digital controller then initiates a digitization of the buffer output voltage and resets the CSA so it is ready for the next charge packet.

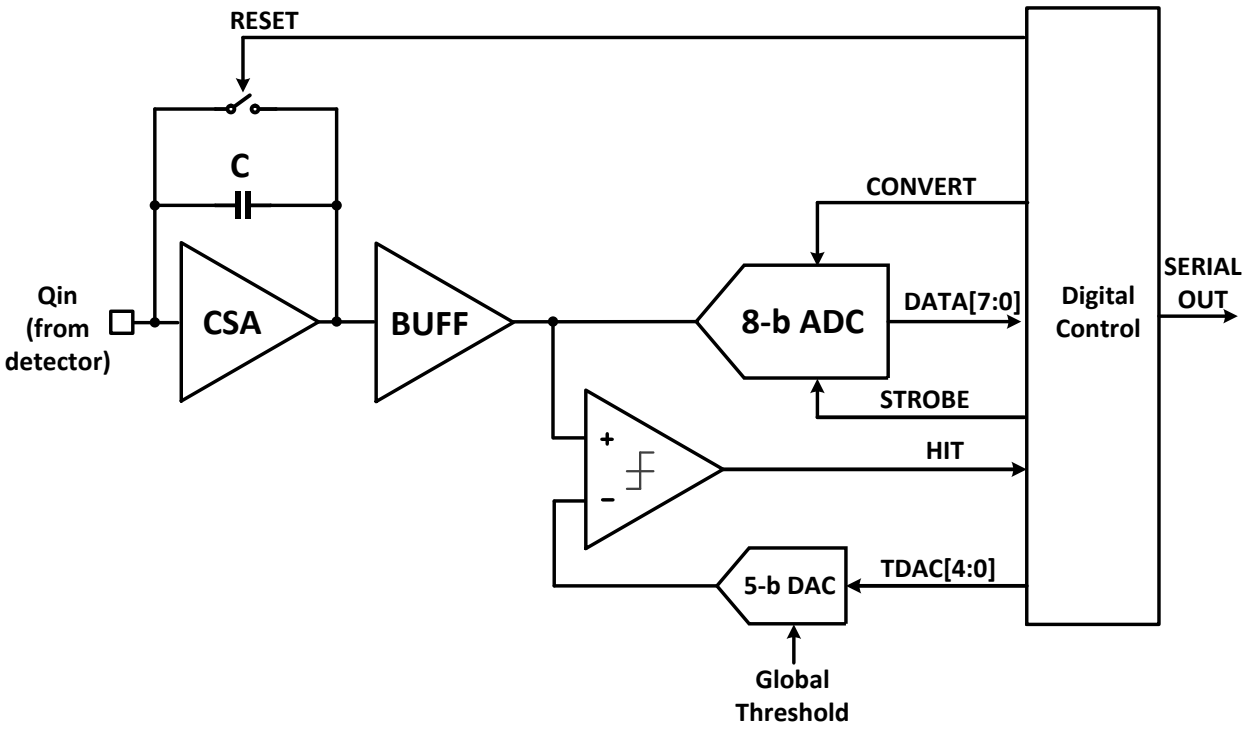


Figure 6. Conceptual block diagram of LArPix-v2b channel.

Various aspects of the data acquisition process are user-programmable as described below. The ADC LSB is determined by

$$LSB = \frac{V_{REF} - V_{CM}}{256}$$

The ADC output is related to its input voltage as follows:

$$D_{out} = \frac{V_{in} - V_{CM}}{V_{REF} - V_{CM}} 2^N$$

V_{REF} sets the ADC maximum input and V_{CM} sets the ADC minimum signal. V_{CM} must be set to above 300 mV for the ADC to function. Set (V_{REF} – V_{CM}) to the expected range of the input to maximize dynamic range.

VCM is set via **vcm_dac[7:0]** and VREF is set **vref_dac[7:0]**. In both cases the reference voltage is set to the DAC value divided by 256 times the power supply voltage.

7.2 Configuration Register Map

The configuration register map for LArPix-v2b is shown in Table 9[2].

Signal Name	Description	Reg	# of bits	Default	Comment
pixel_trim_dac	Pixel-level threshold DAC for channel X	0[4:0] – 63[4:0]	5	5'b0	Each channel has independent trim setting
threshold_global	Global threshold DAC (shared across all channels)	64	8	8'hFF	All channels disabled at power up
csa_gain	Selects CSA gain	65[0]	1	1'b0	0 → 4 mV/ke$^-$ 1 → 2 mV/ke$^-$
csa_bypass_enable	Enables bypass of CSA and injects signal from ADC_TEST directly into selected ADCs	65[1]	1	1'b0	
bypass_caps_en	Enable internal supply bypassing	65[2]	1	1'b1	Active high (default is internal bypassing enabled)
csa_enable	High to enabled CSA (otherwise leave in hard reset)	66-73	64	64'b1	Power up with all CSAs in hard reset. Ex: 73[7] = channel 63
ibias_tdac	Threshold DAC bias current	74[3:0]	4	4'b1000	LSB = IREF/30 (see below)
ibias_comp	Discriminator bias current	75[3:0]	4	4'b1000	LSB = IREF/30 (see below)
ibias_buffer	Channel Buffer bias	76[3:0]	4	4'b1000	LSB = IREF/10

[2] The Defaults are based on Verilog HDL notation. The number before the ' indicates the number of bits or hex digits in the number, the letter b or h after ' indicates whether the number is in binary or hexadecimal, and a single value after the letter indicates all bits or hex digits are identical. A single 0 or 1 means the bit is repeated. For example, 8'b0 means 0000_0000 (or eight individual bits set to zero) and 8'h03 means the 8-bit number 0000_0011.

	current				
ibias_csa	CSA bias current	77[3:0]	4	4'b1000	LSB = 4*IREF/25
ibias_vref_buffer	VREF buffer bias current	78[3:0]	4	4'b1000	LSB = IREF/15 (active low): ibias = (15-code)*LSB
ibias_vcm_buffer	VCM buffer bias current	79[3:0]	4	4'b1000	LSB = 2*IREF/15 (active low): ibias = (15-code)*LSB
ibias_tpulse	CSA pulser bias current	80[3:0]	4	4'b0101	LSB = IREF/10
ref_current_trim	Trim for IREF (primary reference current)	81[4:0]	5	5'b1_0000	IREF = 30 µA nominal (IREF = 1.5*I_READBACK). @warm: IREF = (24.5 + 0.5*code) µA @cold: IREF = (22.75 + 0.4*code) µA
override_ref	High to enable bypassing reference block with external bias resistor	81[5]	1	1'b0	Allows use of external resistor to ground instead of internal reference to generate IREF. @warm: RBIAS = 32 kΩ @cold: RBIAS = 26.7 kΩ (want 800 mV at pin)
ref_kickstart	Kickstarts reference	81[6]	1	1'b0	Only use if reference does not start up automatically. Return to 0 immediately.
vref_dac	DAC to set VREF for ADC	82	8	8'hDB	VREF = code*VDDA/256
vcm_dac	DAC to set VCM for ADC	83	8	8'h4D	VCM = code*VDDA/256
csa_bypass_select	Selects ADC channel(s)	84[0] –	64	64'b0	csa_bypass_enable must = 1

	for test.	91[7]			for this configuration to have effect. 84[0] is channel 0.
csa_monitor_select	Selects channel to connect to analog monitor bus	92[0] – 99[7]	64	64'b0	Active high (set to 1 to select channel). 92[0] is channel 0. Only select one channel at a time.
csa_testpulse_enable	Selects channel(s) to connect to CSA test pulse	100[0] – 107[7]	64	64'b1	Active low (Should be set to high in normal operation). 100[0] is channel 0.
csa_testpulse_dac[7:0]	Set test pulse amplitude	108	8	8'b0	see section on test pulse
current_monitor_bank0	Current monitor for bank 0. Current routed to ANA_MONITOR pin	109[3:0]	4	4'b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
current_monitor_bank1	Current monitor for bank 1. Current routed to ANA_MONITOR pin	110[3:0]	4	4'b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
current_monitor_bank2	Current monitor for bank 2. Current routed to ANA_MONITOR pin	111[3:0]	4	4'b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp bit[0]: ibias_tdac
current_monitor_bank3	Current monitor for bank 3. Current routed to ANA_MONITOR pin	112[3:0]	4	4'b0	bit[3]: ibias_csa bit[2]: ibias_buffer bit[1]: ibias_comp

					bit[0]: ibias_tdac
voltage_monitor_bank0	Voltage monitor for bank 0. Voltage routed to ANA_MONITOR pin	113[2:0]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_bank1	Voltage monitor for bank 1. Voltage routed to ANA_MONITOR pin	114[2:0]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_bank2	Voltage monitor for bank 2. Voltage routed to ANA_MONITOR pin	115[2:0]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_bank3	Voltage monitor for bank 3. Voltage routed to ANA_MONITOR pin	116[2:0]	3	3'b0	bit[2]: vcm_adc bit[1]: vref_adc bit[0]: vin_csa
voltage_monitor_refgen	Voltage monitor for reference generation block. Voltage routed to ANA_MONITOR pin. The "set" voltages are raw DAC outputs (before buffering, used to verify DAC functionality)	117[2:0]	8	8'b0	bit[7]: vref_adc bit[6]: vref_adc_set bit[5]: vcm_adc bit[4]: vcm_adc_set bit[3]: csa_tpulse_internal bit[2]: csa_tpulse bit[1]: csa_thresh_global bit[0]: biasp1
digital_monitor_enable	High to enable digital monitor port	118[0]	1	1'b0	
digital_monitor_sel	Selects digital monitor	118[4:1	4	4'b0	10: sync_timestamp

ect	output signal	]			9: reset_n_config_syn 8: cross_trigger 7: external_trigger 6: periodic_reset 5: periodic_trigger 4: triggered_natural 3: csa_reset 2: sample 1: comp 0: hit
digital_monitor_chan	Selects the channel to observe with monitor	119[5:0]	6	6'b0	
adc_hold_delay	Number of clock cycles for sampling in ADC (i.e. how many cycles to wait before transitioning the SAR into hold mode	120[7:0] - 121[7:0]	16	16'0	In normal operation, the SAR is in sampling mode for at least three clock cycles (track/hold operation). ADC hold delay adds additional clock cycles for sampling at the expense of dead time. High order bits in register 121.
chip_id	Unique ID used to address and identify chip	122	8	8'h01	Default ID is 1. Broadcast ID is 255 (all chips respond to commands with Chip ID = 255, regardless of their individual IDs).
tx_dynamic_powerdown_cycles	Number of cycles to keep TX powered down	123[0]	1	1'b0	0 → TX always powered up when enabled. 1 → TX powered down when not transmitting

load_config_defaults	Active high.	123[1]	1	1'b0	Must be set low again immediately to allow further interaction with registers.
enable_FIFO_diagnostics	High to embed FIFO counter information in output word	123[2]	1	1'b0	When active, bits [43:32] are 12-bit shared FIFO counter value at time event is latched, bits [45:44] are 2-bit local FIFO counter, and bits [47:46] are 0.
enable_tx_dynamic_powerdown	High to prevent and enable TX from powering down when not transmitting	123[0]	1	1'b0	0 → TX always powered up when enabled. 1 → TX powered down when not transmitting
clk_ctrl	Configures ADC (channel) speed	123[4:3]	2	2'b0	Divide ratio between ADC and TX: 00 → 2 01 → 4 10 → 8 11 → 16
tx_dynamic_powerdown_cycles	Number of cycles to keep TX powered down	123[7:5]	1	3'b010	Actual delay is tx_dynamic_powerdown_cycles + 1
enable_PISO_upstream	Enable different PISOs for upstream operation. (For data originating in FPGA)	124[3:0]	4	4'b0	bit[3]: MIS03 bit[2]: PISO2 bit[1]: PISO1 bit[0]: PISO0
enable_PISO_downstream	Enable different PISOs for downstream operation (for data destined for FPGA)	125[3:0]	4	4'b0	bit[3]: MIS03 bit[2]: PISO2

					bit[1]: PISO1 bit[0]: PISO0
enable_POSI	Enable different POSIs	126[3:0]	4	4'b1	bit[3]: M0SI3 bit[2]: POSI2 bit[1]: POSI1 bit[0]: POSI0
test_mode_uart0	Puts UART0 into test mode	127[1:0]	2	2'b0	00 → no test 01 → PRBS5 test 10 → timestamp test 11 → no test
test_mode_uart1	Puts UART1 into test mode	127[3:2]	2	2'b0	00 → no test 01 → PRBS5 test 10 → timestamp test 11 → no test
test_mode_uart2	Puts UART2 into test mode	127[5:4]	2	2'b0	00 → no test 01 → PRBS5 test 10 → timestamp test 11 → no test
test_mode_uart3	Puts UART3 into test mode	127[7:6]	2	2'b0	00 → no test 01 → PRBS5 test 10 → timestamp test 11 → no test
enable_cross_trigger	Enable LArPix-v2b cross trigger	128[0]	1	1'b0	All unmasked channels read out when any channel hit

enable_periodic_reset	Channels reset periodically even if no hit. All channels reset simultaneously.	128[1]	1	1'b0	Number of clocks between resets specified in periodic_reset_cycles (registers 163-165)
enable_rolling_periodic_reset	Channel is reset periodically even if no hit. Channels reset in a rolling fashion (one at a time)	128[2]	1	1'b0	Number of clocks between resets specified in periodic_reset_cycles (registers 163-165)
enable_periodic_trigger	Channels trigger periodically even if no hit. All unmasked channels trigger simultaneously.	128[3]	1	1'b0	Number of clocks between triggers specified in periodic_trigger_cycles (registers 166-169)
enable_rolling_periodic_trigger	Channel is triggered periodically even if no hit. Channels triggered in a rolling fashion (one at a time)	128[4]	1	1'b0	Number of clocks between triggers specified in periodic_reset_cycles (registers 163-165)
enable_periodic_trigger_veto	High to enable the hit signal to veto the periodic trigger	128[5]	1	1'b1	If high, periodic trigger is skipped if channel servicing a hit
enable_hit_veto	High to enforce that discriminator hit is high before channel goes into ADC hold mode	128[6]	1	1'b1	If high, channel is not triggered on transient discriminator hits
shadow_reset_length	Number of clock cycles to hold CSA reset when asserted	129	8	8'b0	OR'd with reset_length (register 170[5:3])
adc_burst_length	Number of ADC conversions per channel hit	130	8	8'b0	Maximum burst length = 255 ADC conversions

channel_mask	High to disable channel	131-138	64	64'b1	131[0] = channel 0
external_trigger_mask	High to disable external trigger for a channel	139-146	64	64'b1	
cross_trigger_mask	High to disable cross trigger for a channel	147-154	64	64'b1	
periodic_trigger_mask	High to disable periodic trigger for a channel	155-162	64	64'b1	
periodic_reset_cycles	Number of clock cycles between periodic resets	163-165	24	24'h00_10_00	
periodic_trigger_cycles	Number of clock cycles between periodic triggers	166-169	32	32'h0	
enable_dynamic_reset	High to enable resetting channel based on ADC data.	170[0]	1	1'b0	Set **adc_burst_length** to maximum desired conversion
enable_min_delta_adc	High to enable mode that resets channel when ADC values have settled	170[1]	1	1'b0	**enable_dynamic_reset** must be low for this bit to have effect
threshold_polarity	High to trigger when ADC above threshold, low to trigger when ADC below threshold	170[2]	1	1'b1	
reset_length	Number of clock cycles to hold CSA reset when asserted	170[5:3]	3	3'b001	Setting this to 0 has same effect as setting to 1
mark_first_packet	Sets MSB of timestamp to '1' for first packet, '0' for	170[6]	1	1'b1	

	other packets				
reset_threshold	In dynamic mode, only reset when ADC is larger than this value	171	8	8'b0	
min_delta_adc	A difference in successive conversions less that this value triggers a channel reset (and ends the ADC burst). This word only has effect if enable_dynamic_reset is high and enable_min_delta_adc is low). This is modulated by threshold polarity.	172	8	8'b0	For example, if threshold polarity == 1, and min_delta_adc == 5, then adc[x] = 200 and adc[x-1] = 197 would trigger a reset, but adc[x] = 200 and adc[x-1] = 203 would not.
digital_threshold	Only put words into FIFO if they are at least this value	173-236	8	8'b0	Word 173 corresponds to channel 0
RESERVED		237-238			Unused in LArPix-v2b
tx_slices0	Number of slices to enable for PISO0 link	239[3:0]	4	4'h8	Active high. Setting tx_slices0 = 4'b0 disables PISO0.
tx_slices1	Number of slices to enable for PISO1 link	239[7:5]	4	4'h8	Active high. Setting tx_slices1 = 4'b0 disables PISO1.
tx_slices2	Number of slices to enable for PISO2 link	240[3:0]	4	4'h8	Active high. Setting tx_slices2 = 4'b0 disables PISO2.
tx_slices3	Number of slices to enable for PISO3 link	240[7:5]	4	4'h8	Active high. Setting tx_slices3 = 4'b0 disables PISO3.
i_tx_diff0	Differential per-slice loop current for PISO0	241[3:0]	4	4'hC	Active low. LSB = IREF/30.

	link				
i_tx_diff1	Differential per-slice loop current for PISO1 link	241[7:4]	4	4'hC	Active low. LSB = IREF/30.
i_tx_diff2	Differential per-slice loop current for PISO2 link	242[3:0]	4	4'hC	Active low. LSB = IREF/30.
i_tx_diff3	Differential per-slice loop current for PISO3 link	242[7:4]	4	4'hC	Active low. LSB = IREF/30.
i_rx0	Receiver bias current for POSI0 link	243[3:0]	4	4'h8	Active Low. LSB = IREF/50.
i_rx1	Receiver bias current for POSI1 link	243[7:4]	4	4'h8	Active Low. LSB = IREF/50.
i_rx2	Receiver bias current for POSI2 link	244[3:0]	4	4'h8	Active Low. LSB = IREF/50.
i_rx3	Receiver bias current for POSI3 link	244[7:4]	4	4'h8	Active Low. LSB = IREF/50.
i_rx_clk	Receiver bias current for clock rx	245[3:0]	4	4'h8	Active Low. LSB = IREF/50.
i_rx_rst	Receiver bias current for reset rx	245[7:4]	4	4'h8	Active Low. LSB = IREF/50.
i_rx_ext_trig	Receiver bias current for external trigger rx	246[3:0]	4	4'h8	Active Low. LSB = IREF/50.
r_term0	Termination resistor for POSI0 link	247[4:0]	4	5'h02	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term0 = 0. Default is 2 kΩ.
r_term1	Termination resistor for POSI1 link	248[4:0]	4	5'h02	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term1 = 0. Default is 2 kΩ.

r_term2	Termination resistor for POSI2 link	249[4:0]	4	5'h02	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term2 = 0. Default is 2 kΩ.
r_term3	Termination resistor for POSI3 link	250[4:0]	4	5'h02	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term3 = 0. Default is 2 kΩ.
r_term_clk	Termination resistor for clock rx	251[4:0]	4	5'h00	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term0 = 0. Default is 2 kΩ.
r_term_reset	Termination resistor for reset rx	252[4:0]	4	5'h00	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term0 = 0. Default is 2 kΩ.
r_term_ext_trig	Termination resistor for external trigger rx	253[4:0]	4	5'h00	Total resistance = 3 kΩ - r_term (see text). Hi-Z when r_term0 = 0. Default is 2 kΩ.
v_cm_lvds_tx0	Common-mode voltage for PISO0 link	254[2:0]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.
v_cm_lvds_tx1	Common-mode voltage for PISO1 link	254[6:4]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.
v_cm_lvds_tx2	Common-mode voltage for PISO2 link	255[2:0]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.
v_cm_lvds_tx3	Common-mode voltage for PISO3 link	255[6:4]	3	3'h5	LSB = VDD/8 (nominal is VDD/2). Active high.

Table 9. LArPix-v2b configuration register map.

7.3 Channel Threshold

The channel threshold consists of two components. A global threshold set by **threshold_global[7:0]** and a per-channel trimmable threshold set by **pixel_trim_dac[4:0]**.

The global threshold value is:

$$Th_{Gl} = \frac{VDDA}{256}\boldsymbol{threshold_global}[\mathbf{7}:\mathbf{0}]$$

At room temperature (300K) each channel threshold is:

$$Th_chan_RT_X = Th_{Gl} + 210\ mV + 1.45mV * \boldsymbol{pixel_trim_dacX}[\mathbf{4}:\mathbf{0}]$$

At 88 K each channel threshold is:

$$Th_chan_88K_X = Th_{Gl} + 465\ mV + 2.34mV * \boldsymbol{pixel_trim_dacX}[\mathbf{4}:\mathbf{0}]$$

Please keep in mind the absolute accuracy of the thresholds is greatly affected by mismatch (especially at cold).

8 Digital I/O

LArPix-v2b includes a programmable current mode digital I/O to reduce crosstalk. A block diagram of the system is shown in Figure 7. The transmitted bit is converted to a differential current at the transmitter chip and sent to the receiver chip. At the receiver chip, the differential current is converted to a differential voltage by a termination resistor (which may be internal or external to the receiving chip). In the receiving chip, an amplifier senses the differential voltage and outputs a logic-level CMOS bit.

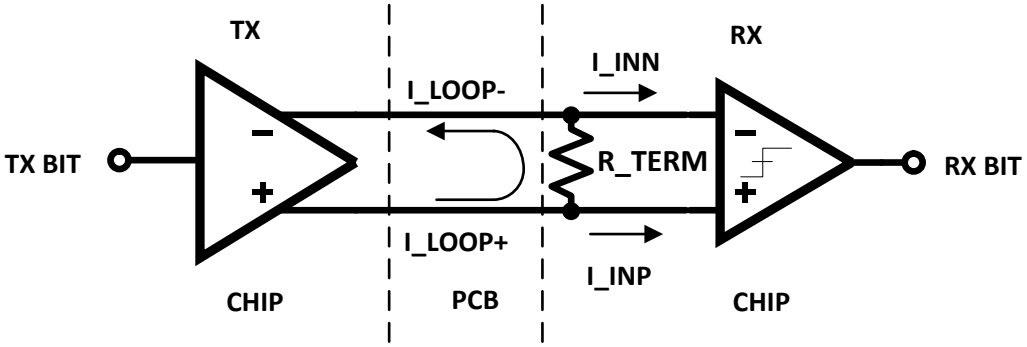


Figure 7. LArPix-v2b differential digital I/O.

The common-mode voltage (the average voltage of the two parts of the loop) and the loop current can be adjusted on the TX side and the termination resistance and receiver bias current can be adjusted on the RX side. The loop current can be adjusted in two ways. The transmitter is implemented by 15 parallel

transmit modules. For large changes in loop current, additional transmit slices can be added or removed from operation. For fine tuning, the raw per-slice current can also be adjusted. This ensures proper biasing at cold even over a wide range of possible loop currents.

8.1 Digital I/O TX Configuration

The Digital I/O transmitter can be configured by adjusting the number of parallel slices engaged, the loop current contributed by each slice, and the common-mode voltage of the transmission.

There are 15 parallel slices. The number of slices connected to the output can be adjusted by changing **tx_slicesX[3:0]** (where, for instance, **tx_slices0[3:0]** controls the number of slices contributing to the PISO0 loop current.

The amount of current contributed by each slice is controlled by the active low control **i_tx_diffX[3:0]**.

The loop current in the link is defined by the following equation:

i_loop_tx = **tx_slicesX[3:0]** * (15- **i_tx_diffx[3:0]**)*i_ref/30.

i_ref is the Primary bias current that is set by the on-chip biasing module (see below for information on adjusting i_ref).

The common-mode voltage of the transmitter is set by adjusting the active high control **v_cm_lvds_txX[2:0**]. The LSB of the DAC that sets of common-mode voltage is VDD/8.

8.2 Digital I/O RX Configuration

The Digital I/O receiver can be configured by adjusting its bias current and its internal termination resistor.

The bias current in the receiver is adjusted using the active low control **i_rxX[3:0]**. The value of the bias current is:

i_bias_rx = (15 – **i_rxX[3:0]**)* i_ref/50.

i_ref is the primary current set but the on-chip biasing module (see below for information on adjusting i_ref).

The termination resistor is set by adjusting **r_termX[4:0]**. The termination resistor is implemented using internal 2.5 kΩ resistors. So, the termination resistor is approximately 2.5 kΩ/**r_termX[4:0]**. When **r_termX[4:0]** is set to zero, then the receiver is high impedance and the channel must be terminated external to LArPix-v2b.

8.3 Digital I/O Configuration Summary

The settings to adjust the various controls available for the Digital I/O are summarized in Table 10. The "X" in each control name indicates which PISO/POSI port it is affecting.

Configuration Word Name	Adjustment	Bias Transfer Function LSB
tx_slicesX[3:0]	Number of parallel TX slices engaged	LSB = unit slice
i_tx_diffX[3:0]	Loop current contributed per slice	LSB = i_ref/30 (active low)
v_cm_lvds_txX[2:0]	Transmitter common-mode output voltage	LSD = VDD/8
i_rxX[3:0]	Bias current for receiving amplifier	LSB = i_ref/50 (active low)
r_termX[4:0]	Input termination resistor for receiving amplifier	LSB = 2.5 kΩ resistor. Word indicates number of 2.5 kΩ unit devices that comprise termination resistor.

Table 10. Digital I/O Configuration Summary.

9 Hydra Network

9.1 Introduction to Hydra

The Hydra I/O is the organizational architecture for connecting multiple LArPix-v2b chips into a network. Because of the large number of LArPix-v2b chips used in a practical system, it will be important to limit the physical lines of communications (both to reduce the number of cryostat penetrations and to simplify PCB design). The approaches used in LArPix-v1 and LArPix-v2b are contrasted in Figure 8. LArPix-

v1 had a single UART per chip and used a simple daisy chaining approach to satisfy these requirements. Each LArPix-v1 ASIC was connected to its two nearest neighbors in one dimension and all the ASICs were organized in a series of columns. All data would be forward to the next downstream ASIC until it would finally exit the cryostat. As is, this approach is simple and easy to implement but it not scalable since a single ASIC failure would make it impossible to communicate with all ASICs upstream of the failed ASIC on the daisy chain.

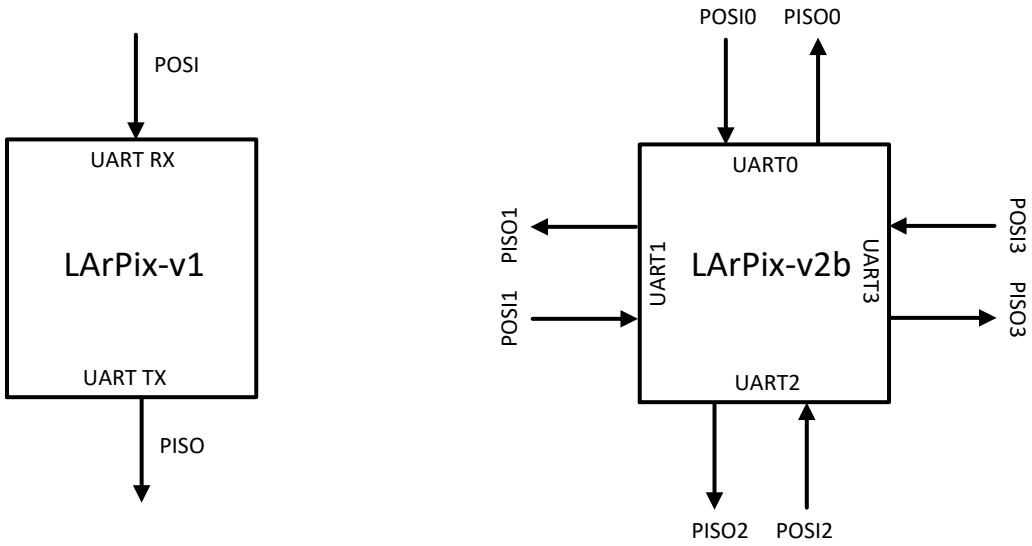


Figure 8. Comparison of available UARTs between LArPix-v1 and LArPix-v2b (figure adapted from original by Dan Dwyer).

To circumvent the reliability problems of a daisy chain approach to LArPix interconnection a new architecture called Hydra is employed. As seen in Figure 8, LArPix-v2b has four on-board UARTs and each LArPix-v2b is connected to its nearest neighbors on all four sides, and which side is the dominant direction to shift data toward the outside is configurable. Therefore, the organization is not constrained to be daisy chains and 2-D networks of arbitrary complexity can be constructed in an ad hoc way. This means single failed ASICs can be easily routed around (which was problematic from an implementation standpoint in the daisy chain approach).

In Hydra there are downstream and upstream channels. Upstream channels are used to send configuration commands through the network. Downstream channels are used to return data and requested configuration data to the controller. Downstream and upstream channels use the same PISO ports so the difference is logical rather than physical. Downstream packets are marked as such to differentiate them from upstream packets.

While Hydra provides the means by which large numbers of LArPix-v2b chips and collaborate, each individual LArPix-v2b ASIC knows nothing about Hydra I/O and is only be able to power up and down

UARTs and respond to commands that match its Chip ID. It will be up to the configuration script author to ensure that pathological things (like configuration infinite loops) do not occur. Caveat Emptor.

9.2 Chip ID

Each individual ASIC in a Hydra network requires an individual ID that allows it to be addressed. LArPix ASICs will only respond to commands that are addressed to their chip ID or to the broadcast chip ID of 255. All commands sent to Chip ID 255 will be processed by all ASICs. To change the chip ID of an individual LArPix ASIC, update the value in the register **chip_id[7:0]**.

9.3 Example Hydra Network

An example Hydra network of 25 LArPix-v2b ASICs arranged in a 5-by-5 grid is shown in Figure 9.

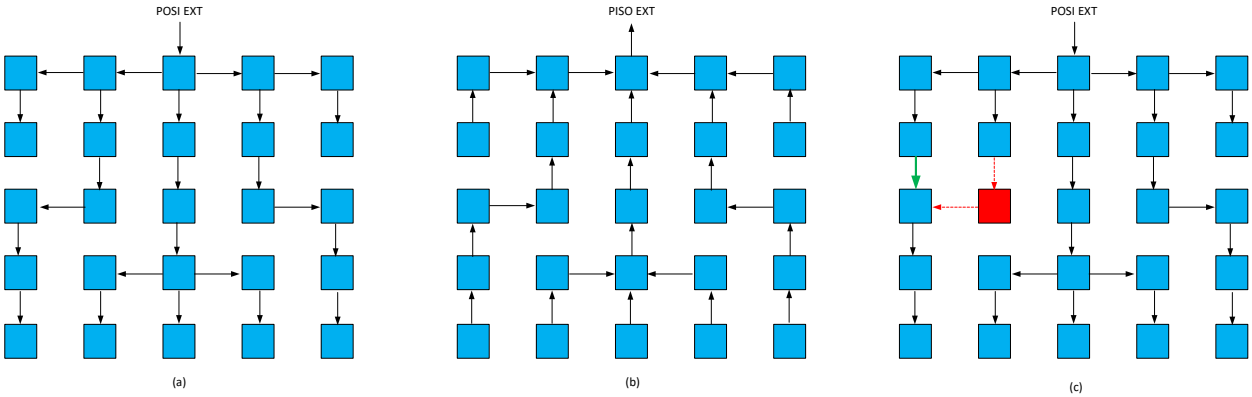


Figure 9. Hydra Network. a) upstream network, b) downstream network, c) upstream network re-routing around damage. (figure adapted from original by Dan Dwyer)

The upstream Hydra network in Figure 9a is used to configure the various LArPix-v2b ASICs in the network. Each chip is given a unique Chip ID and POSI ports are enabled to make an efficient topology to configure a large number of ASICs. The corresponding upstream network is shown in Figure 9b. In the case of an ASIC failure, the Hydra network can be reconfigured to route around the damage. This is shown in Figure 9c. The red ASIC has failed and the red, dotted connections are no longer operable. Without Hydra's ability to reconfigure, not only the red chip would be lost but all downstream ASICs as well would be out of communication. Using Hydra, however, the green connection can be configured by enabling a different POSI port and all of the chips in the network are once again reachable.

9.4 UART Configuration

Each of the four UARTs on LArPix-v2b can be configured independently. Each POSI can be enabled or disabled (although care should be taken to make sure the ASIC can still be addressed) using the configuration bits **enable_POSI[3:0]**, where each configuration bit corresponds to a single POSI output. The PISO ports can be configured as upstream or downstream using the **enable_PISO_upstream[3:0]** and **enable_PISO_downstream[3:0]** bits.

LArPix-v2b includes a pseudo-differential CMOS output to mitigate electromagnetic interference. The differential output can be enabled on a per-PISO basis using the configuration bits **enable_PISO_differential[3:0]**.

The UARTs contain a physical-layer testmode that does not include event building. Each UART PISO port can output pseudo-random data (PRBS5) or a timestamp packet. When in PRBS5 test mode, the UART will continuously transmit two 1's followed by two copies of the PRBS5 word 31'h6B3E3750. When it timestamp mode it will send out the lower 31 bits of the timestamp followed by the complete timestamp. In other words the output of the UART will be {timestamp_32b[31:0],timestamp_32b}. As a reminder, the UART sends data LSB first.

Each UART can be configured for test independently. The test control bits are summarized in Table 11.

Config Word Bits	Function
00	No Test
01	PRBS5 Test
10	Timestamp Test
11	No Test

Table 11. UART Test Modes.

9.5 Hydra Network Bringup

To bring up a Hydra network, during configuration each ASIC will need to:

- Update Chip ID to a unique value (default Chip ID is 1)
- Update configuration registers as desired
- Enable downstream PISO to all configuration read commands

- Verify all registers are correctly configured
- Enable appropriate PISO channel for one desired upstream channel
- Contact next upstream ASIC and configure
- Repeat for each ASIC in desired Hydra network configuration

The basic idea is that only one chip that is actively attached to the network will have Chip ID at 1 at any time. Otherwise, it may become ambiguous which chip is currently under configuration.

As an example, consider the simple four-chip Hydra network in Figure 10. There is a single connection to the outside work (FPGA) that will be used to configure the entire network. At power up, each chip has Chip ID = 1. To configure the network as shown the following commands can be issued from the controlling FPGA (or computer) as seen in Table 12.

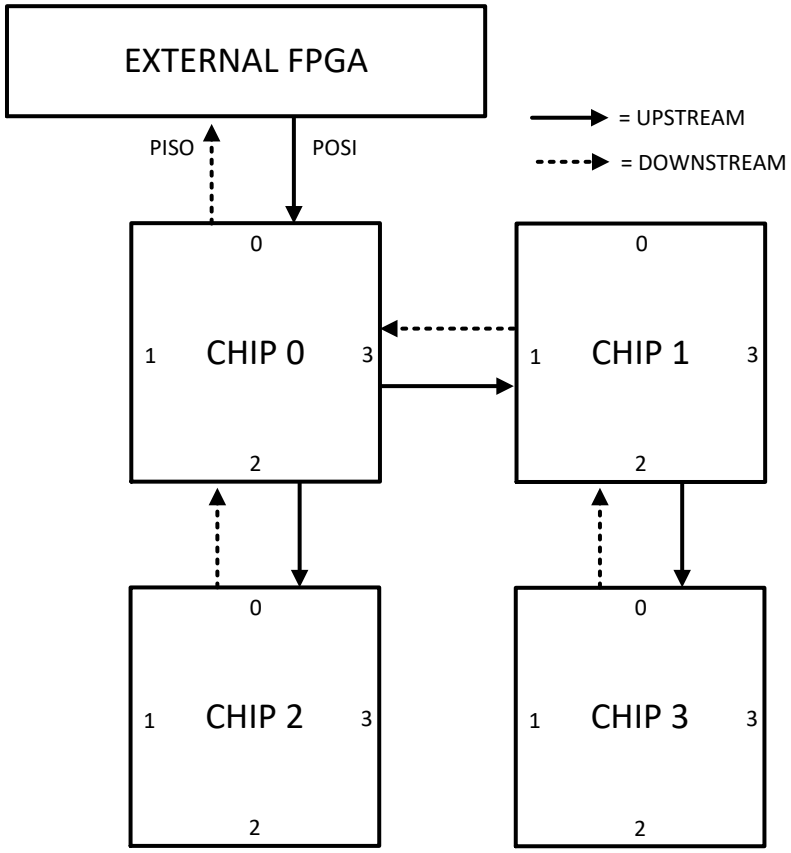


Figure 10. Four-chip Hydra network.

Step	Chip ID	Config Signal Name	Value	Explanation

1	1	chip_id	4	Change Chip ID of Chip 0 from 1 to 4.
2	4	enable_PISO_downstream	1	Enable Chip 0 - PISO0 to return downstream data to FPGA
3	4	enable_PISO_upstream	8	Enable Chip 0 – PISO3 (for writing data to chip 1)
4	1	chip_id	2	Change Chip ID of Chip 1 from 1 to 2.
5	2	enable_PISO_downstream	2	Enable Chip 1 – PISO1 to return downstream data to Chip 0
6	2	enable_PISO_upstream	4	Enable Chip 1 – PISO2 (for writing data to chip 3)
7	1	chip_id	3	Change Chip ID of Chip 3 from 1 to 3.
8	3	enable_PISO_downstream	1	Enable Chip 3 – PISO0 to return downstream data to Chip 1
9	4	enable_PISO_upstream	4	Enable Chip 0 – PISO3 (for writing data to chip 2)
10	1	chip_id	5	Change Chip ID of Chip 3 from 1 to 5.
11	5	enable_PISO_downstream	1	Enable Chip 2 – PISO0 to return downstream data to Chip 0

Table 12. Algorithm to configure example Hydra network.

10 Reset Modes

LArPix-v2b has three reset modes: normal (with user defined reset stretching), periodic, and rolling periodic, and minimum delta ADC. The number of clock cycles the CSA spends in reset is determined by **reset_length[2:0]**. Setting this to 0 or 1 has the same effect (reset is asserted for one clock cycle). Increasing **reset_length[2:0]** has the benefit of making charge removal from the CSA feedback capacitor more ideal.

10.1 Normal Reset Mode

In normal reset mode, the channel is reset as soon as the sampling process has completed.

10.2 Periodic Reset

There may be situations when it is desirable to periodically reset one or more channels. For example, background events will integrate on the front-end output capacitor and will eventually cause a false hit[3]. To mitigate this, the channels can be reset periodically (at the cost of a small about of dead time). To put LArPix-v2b into periodic reset mode, set **enable_periodic_reset** to one. LArPix-v2b contains an internal counter that will trigger a reset after a given number of system clock periods. The internal counter increments every two system clocks. The number of counter increments is given by the 24 bit word **periodic_reset_cycles[23:0]** in registers 163-165. Register 165 holds the MSB byte and register 163 holds the LSB byte. The counter is incremented every four cycles of the primary clock.

Channels can be masked from responding to a periodic reset by setting **periodic_reset_mask[63:0]** to 1 for the desired channels. Note that channels automatically ignore periodic reset commands if they are currently processing a hit.

10.3 Rolling Periodic Reset

By default, all unmasked channels are reset at the same time when the reset counter equals the value in **periodic_reset_cycles[23:0]**. A separate rolling periodic reset mode is accessible by setting **enable_rolling_periodic_trigger** = 1. In this mode, only one channel is reset each time the reset counter equals the value in **periodic_reset_cycles[23:0]**.

10.4 Dynamic Reset

In dynamic reset mode, the channel does not reset until the ADC value is above (or below) the threshold specified in **reset_threshold[7:0]**. Set **threshold_polarity** = 1 for the channel to reset when the current

[3] The false hit rate is expected to be much larger at RT than at LAr temperature because of the increased device leakage (that will be integrated onto the feedback capacitor) at RT.

ADC value is above the threshold and set **threshold_polarity** to 0 for the channel to reset when the ADC value is below the threshold.

10.5 Minimum Delta ADC

In this mode, the channel does not reset until the ADC values have settled to a user-specified amount. TO enable this mode, set **enable_min_delta_adc** to 1. Then, the channel will not reset itself until the difference between two consecutive ADC values is less than the value in **min_delta_adc[7:0]**. To make sure the channel does not hang, the maximum number of conversions the channel will spend waiting is set my **adc_burst_length[7:0]**. Note that **enable_dynamic_reset** must be set to 0 for this mode to function.

11 Trigger Modes

LArPix-v2b has a variety of trigger modes to allow increased flexibility during ASIC and system evaluation. Each of the trigger modes has a corresponding mask that is described below.

11.1 Hit Veto

There is a hit veto function which can be important to set correctly if various trigger modes are desired. The configuration bit **enable_hit_veto** = 1 requires that the discriminator output is high before the channel will go into hold mode and begin the digitization process. This can be useful as a noise filter in Normal Trigger mode but will disable other trigger modes. Therefore, be sure to disable the **enable_hit_veto** if trigger modes different from normal are desired.

11.2 Normal Trigger

In normal mode, a LArPix-v2b channel triggers when the input signal exceeds the threshold of the discriminator.

11.3 Cross-Trigger

In cross-trigger mode all un-masked channels (see section on masking below) of LArPix-2 are read out when any channels receive a hit. To put LArPix-v2b in cross-trigger mode, set **enable_cross_trigger** to 1. Be advised that in the presence of significant channel occupancy cross-trigger mode could lead to FIFO overflow.

11.4 External Trigger

LArPix-v2b can also trigger on an external signal applied to the EXT_TRIG pin. In this case, all channels configured to respond to the external trigger signal (i.e. unmasked) will trigger when EXT_TRIG is asserted, regardless of the input level.

11.5 Periodic Trigger

LArPix-v2b can trigger passed on a periodic trigger in a manner analogous to the Periodic Reset. To enable the periodic trigger functionality, assert **enable_periodic_trigger** and be sure the channels of interest are configured to respond. There is also a dedicated veto logic for the periodic trigger which can be configured by setting **enable_periodic_trigger_veto** to 1. When asserted, the otherwise properly configured channel will not respond to a periodic trigger when it is processing a natural hit.

The number of cycles between periodic triggers is set by **periodic_trigger_cycles[63:0]**.

11.6 Rolling Periodic Trigger

Similar to the rolling periodic reset, if **enable_rolling_periodic_trigger** is set to 1 then only one channel will trigger each time the periodic trigger is asserted.

11.7 ADC Burst

In some situations it may be advantageous to take multiple ADC measurements before resetting the channel. For example, it may be more desirable to digitize the waveform as charge integrates in the analog front end rather than taking a snapshot and then resetting. Set **adc_burst_length[7:0]** to the number of additional conversions done per channel hit. After **adc_burst_length[7:0]** conversions, the channel will be reset. Setting **adc_burst_length[7:0]** to either 0x0 or 0x1 will give one conversion. ADC Burst is in effect for all trigger modes and sets a ceiling on the number of ADC conversions generated by a hit.

12 Masks

LArPix-v2b contains various masks configure which channels will respond to various trigger and reset modes.

12.1 Channel Mask

To mask off hot channels (or inhibit the response of select channels for other reasons) set the bits of **channel_mask[63:0]** corresponding to desired masked channels to one. For example, 0xFF000000F will mask out channels 24 through 31 and 0 through 3. Any hits on the masked channels will be ignored.

Important: The channel mask takes priority over all other masks. If a channel is masked using the channel mask it will not respond to any stimulus.

12.2 External Trigger Mask

It may be desirable to inhibit certain channels from responding to an external trigger. To do so, set the bits of **external_trigger_mask[63:0]** corresponding to desired masked channels to one. For example, 0xF will inhibit channels 0 through 3 from responding to an external trigger. All other channels will initiate an ADC conversion upon receiving an external trigger.

12.3 Cross Trigger Mask

It may be desirable to inhibit certain channels from responding to a cross trigger. To do so, set the bits of **cross_trigger_mask[63:0]** corresponding to desired masked channels to one. For example, 0xF will inhibit channels 0 through 3 from responding to a cross trigger. All other channels will initiate an ADC conversion upon receiving a cross trigger.

12.4 Periodic Trigger Mask

It may be desirable to inhibit certain channels from responding to a periodic trigger. To do so, set the bits of **periodic_trigger_mask[63:0]** corresponding to desired masked channels to one. For example, 0xF will inhibit channels 0 through 3 from responding to a periodic trigger. All other channels will initiate an ADC conversion upon receiving a periodic trigger.

13 Digital Threshold

Depending on operational conditions, it may be known that ADC values below a certain value are due to noise hits and are not valid data. Therefore, LArPix-v2b has a digital threshold functionality that inhibits on a per-channel basis the placing of events in the FIFO if the ADC is below the digital threshold. This has the positive impact of reducing power and pressure on the FIFO. Each channel has an independent **digital_threshold[7:0]**. Register 173 corresponds to Channel 0 and Reg 236 corresponds to Channel 63.

14 Monitors

LArPix-v2b has two monitor busses, one analog and one digital. This enables various internal signals to be observed on the bench during evaluation of the prototype. The CSA monitor and Analog monitor use the same physical bus so care must be taken not to short them out (this is not enforced in hardware).

14.1 CSA Monitor

The analog output of each channel can be monitored using the analog monitor bus. The command is active high, so to connect a channel to the analog monitor bus, set (i.e. set to high) the corresponding bit of **csa_monitor_select[63:0]**. For example, to connect the output of channel 7's analog front end to

the analog monitor bus, set **csa_monitor_select[63:0]** to 0x8. Only one channel at a time should be connected to the analog monitor bus across all chips connected to a single bus. This is not enforced in hardware so care should be taken to avoid shorts made by connecting multiple channels to the analog monitor bus.

14.2 Analog Monitor

The analog monitor is used to view various internal signals through the ANA_MONITOR pin. The analog monitor can be used to monitor either internal currents or voltages. The currents are sourced by LArPix-v2b so if monitoring a current there must be a resistor to ground from the ANA_MONITOR pin. Ideally the test board will have a switch or jumper to allow monitoring of both currents and voltages.

The biasing is split into quadrants in the chip and the bias voltages and currents of the different quadrants can be monitored to see if they are consistent across the chip. The analog monitor bits are all one-hot so setting any of them to 1 will make the connection. Therefore, care must be taken to avoid shorting the bus (it is not enforced in hardware). The signals available for monitoring are summarized in Table 13.

Configuration name	Bank Monitored	Signal Monitored	Voltage / Current
current_monitor_bank0[3:0]	0	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current
current_monitor_bank1[3:0]	1	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current
current_monitor_bank2[3:0]	2	bit[3] --> ibias_csa bit[2] --> ibias_buffer	current

		bit[1] --> ibias_comp bit[0] --> ibias_tdac	
current_monitor_bank3[3:0]	3	bit[3] --> ibias_csa bit[2] --> ibias_buffer bit[1] --> ibias_comp bit[0] --> ibias_tdac	current
voltage_monitor_bank0[2:0]	0	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_bank1[2:0]	1	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_bank2[2:0]	2	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_bank3[2:0]	3	bit[2] --> vin_csa bit[1] --> vref_adc bit[0] --> vcm_adc	voltage
voltage_monitor_refgen[2:0]	N/A	bit[0] --> biasp1 bit[1] → csa_glob_thresh bit[2] --> csa_tpulse bit[3] --> csa_tpulse_int bit[4] --> vcm_adc_set bit[5] --> vcm_adc	voltage

		bit[6] --> vref_adc_set bit[7] --> vref_adc	

Table 13. Analog Monitor Settings.

14.3 Digital Monitor

The digital monitor is used to observe various internal digital signals at the DIG_MONITOR pin. Many of the signals are per-channel so to operate the digital monitor it must first be enabled, and then the channel as well as the signal selected.

To enable the digital monitor, set **digital_monitor_enable** to 1. The selected channel is chosen thorugh **digital_monitor_chan[5:0]**. The signal to monitor can be determined by **digital_monitor_select[3:0]**. The decoding of **digital_monitor_select[3:0]** is summarized in Table 14.

digital_monitor_select[3:0] value	**Digital Monitor Output Signal**
0	hit
1	comp
2	sample
3	csa_reset
4	triggered_natural
5	periodic_trigger
6	periodic_reset
7	external_trigger
8	cross_trigger
9	reset_n_config_sync
10	sync_timestamp

Table 14. Digital Monitor Settings.

15 Biasing

The bias points of many of the amplifiers in LArPix-v2b are programmable. All trimming currents reference the Primary bias current.

15.1 Primary Bias

The Primary bias current is generated by an internal vt-referenced current source that is independent of the power supply voltage to first order. The current is trimmed using **ref_current_trim[4:0]**. The current is monitored through the I_MONITOR pad (LArPix-v2b sources the current). Adjust **ref_current_trim[4:0]** until the readback current i_readback is equal to 20 µA. The internal Primary current i_ref is equal to 1.5 * i_readback.

At **300 K**, the primary current i_ref is:

$$i_{ref} = 24.5\ \mu A + 0.5\ \mu A * ref_current_trim[4:0]$$

At **88 K**, the primary_current i_ref is:

$$i_{ref} = 22.75\ \mu A + 0.4\ \mu A * ref_current_trim[4:0]$$

The internal reference current generator uses a startup circuit to ensure it settles to the desired operating point. In case the startup circuit fails to start at cold, the operator can set **ref_kickstart** to 1 to force the operating point and then return **ref_kickstart** to 0 to allow normal operation.

In case the internal reference current generator still fails to operate, an external resistor can be connected from the R_BYPASS pad to ground if necessary. In this mode, **override_ref** should be set to 1. At 300 K the external resistor value should be 32 kΩ and at 88 K it should be 26.7 kΩ. The voltage can be monitored at the pad to ensure 30 µA is flowing. When not used, R_BYPASS should be connected to VDDA.

15.2 Channel Biasing

Various channel blocks can have their bias points set independently. All the programming settings are referenced to the Primary internal bias current i_ref. The bias programming is summarized in Table 15.

Configuration Name	Block Name	Bias Transfer Function LSB
ibias_tdac[3:0]	Threshold DAC	LSB = i_ref/30
ibias_comp[3:0]	Discriminator	LSB = i_ref/30
ibias_buffer[3:0]	A/B Buffer	LSB = i_ref/10
ibias_csa[3:0]	CSA	LSB = 4*i_ref/25
ibias_vref_buffer[3:0]	VREF Buffer	LSB = i_ref/15 (control is active low, so ibias = (15-code)*i_ref/15)
ibias_vcm_buffer[3:0]	VCM Buffer	LSB = 2*i_ref/15 (control is active low, so ibias = (15-code)*2*i_ref/15)
ibias_tpulse[3:0]	CSA Pulser	LSB = i_ref/10

Table 15. Bias Current Programming.

16 Additional Modes and Settings

LArPix-v2b includes a number of functional settings and modes (for both normal operation and test) that are accessible through the Configuration Register Map.

16.1 Front-End Gain

The analog front end has two gain settings. If **csa_gain** is set to 0, the nominal front-end gain is 4 µV/e-. If **csa_gain** is set to 1, the nominal front-end gain is 2 µV/e-.

16.2 Front-End Bypass (to test ADC)

To enable the independent evaluation of the channel ADC, the front-end can be bypassed. Bypassing the front-end is a two-step process. First, the bypass channel must be enabled by setting **csa_bypass_enable** to 1. Then, the channels to bypass (i.e., the channels to which the test voltage pin should be applied) are selected by setting each bit corresponding to each channel of **csa_bypass_select[31:0]**. For example, to connect the test voltage pin (ADC_TEST_IN) to the ADCs in channel 31 and in channels 0 through 4, **csa_bypass_select[31:0]** should be set to 0x8000001F. In this mode either a pulse on the external trigger pin or a test pulse of sufficient amplitude to cause a discriminator hit is required to initiate a conversion.

16.3 Power Supply Bypass Select

LArPix-v2b includes power supply bypassing in the analog front end to reduce transient interference. This bypassing can be disabled by the user if it causes ringing in the power supply network. To enable power supply bypassing, set **bypass_caps_enable** to 1. To disable, set it to 0. The bypassing is disabled by default.

16.4 Analog Test Pulse

LArPix-v2b includes an analog test pulse generator which can be used to characterize the channels. A conceptual diagram of the test pulse generator is shown in Figure 11.

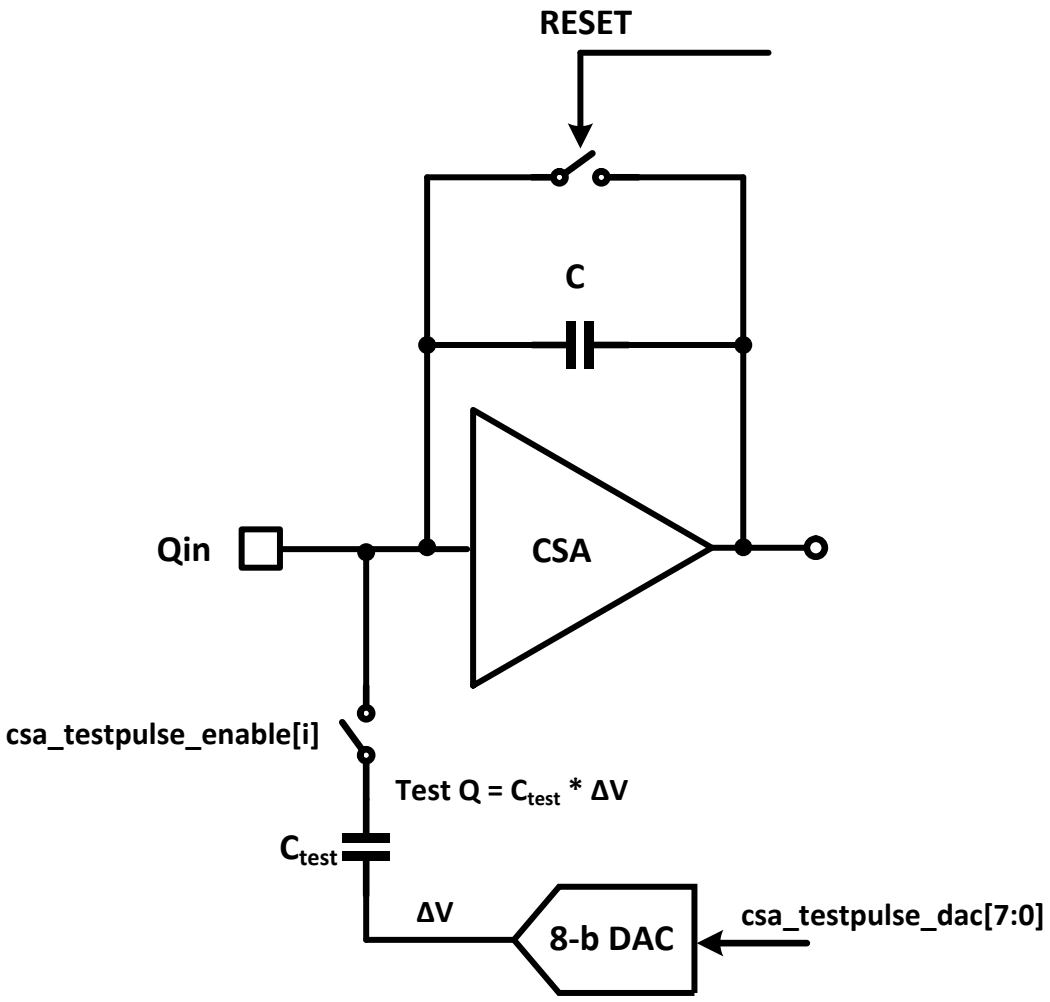


Figure 11. Internal analog test concept.

A channel-under-test is selected by closing the switch labeled **csa_testpulse_enable[i]**. The 8-b test DAC is then updated by changing the **csa_testpulse_dac[7:0]** word. This creates a voltage step ΔV that is coupled to the input of the analog front end by the test capacitor C_{test}. The voltage response of the channel can then be interpreted to extract the gain of the channel. In LArPix, C_{test} is approximately 40 fF. Because C_{test} is approximately 0.1% the size of the expected input capacitance of the channel, the input capacitance will not affect the input charge and the CSA analog output will be

$$V_{out} = \frac{\Delta V C_{test}}{C}$$

where C is the feedback capacitance of the CSA.

The command **csa_testpulse_enable[63:0]** is active low. Therefore, to create a test pulse, first select the channels of interest by setting their corresponding bits in **csa_testpulse_enable[63:0]** to zero**.** Then, update the test DAC by setting the desired value to **csa_testpulse_dac[7:0]**.

The LSB of the test pulse DAC is given by the analog power supply voltage divided by 256. So, if the **csa_testpulse_dac[7:0]** is 128 and the analog power supply is 1.8 V, ΔV would be (1.8V/256) * 128 = 900 mV.

16.5 Sample Stretching

The ADC in LArPix-v2b works by sampling the analog input onto the ADC capacitance and then applying the successive approximation (binary search) algorithm to converge on the closet digital representation of the sampled analog signal. The ADC capacitance is significant and therefore multiple clock cycles may be required to adequately settle depending on the system clock rate. The configuration word **adc_hold_delay[3:0]** determines how many system clock periods the ADC will spend sampling the data. For example, setting **adc_hold_delay[3:0]** to 0xF will result in 15 system clock periods allocated to sampling. Be advised that this will affect channel throughput. Settings zero and one both correspond to one system clock period for sampling.

17 Data Format

The following data is included in each 64-bit event:

1. 2-bit packet declaration (is it a data packet, a test packet, or a configuration packet?)
2. 8-bit chip ID (allowing up to 256 chips to be addressed as a unit)
3. 6-bit channel ID (allowing up to a 64-channel ASIC)
4. 32-bit timestamp
5. 8-bit data payload word
6. Trigger Type (Normal, External, Cross, or Periodic)
7. FIFO half full and FIFO full flags (both shared FIFO and local FIFO)
8. Marker to indicate if a packet is upstream or downstream
9. Parity Bit (LArPix-v2b implements odd parity).

In addition, several MSBs of the timestamp can be replaced with the current FIFO depth, if desired.

An event is defined as either a hit in any channel or a configuration request. Because the time between when a hit occurs and when it is transmitted off chip depends on a variety of factors, it is not deterministic. Therefore, LArPix-v2b includes a 32-bit timestamp. Assuming a 10 MHz system clock, for

instance, a 32-bit time stamp will cover a range of over seven minutes. Additional timestamping can be implemented at the FPGA level.

17.1 Data Packet

The LArPix-v2b event output format (data mode) is shown in Table 16. This format is operative when the Packet Declaration bits are 01.

Bit Range	Contents	Comment
[1:0]	Packet Declaration	**00: data** 01: not used 10: configuration write 11: configuration read
[9:2]	Chip ID	Each chip in a Hydra Network needs a unique ID
[15:10]	Channel ID	64-channel chip supported
[47:16]	Time Stamp	32-bit time stamp
[55:48]	Data Word	8-bit ADC data word
[57:56]	Trigger Type	00: Normal, 01: External, 10: Cross, 11: Periodic
[59:58]	Local FIFO Status	Bit 58: FIFO half full, Bit 59: FIFO full
[61:60]	Shared FIFO Status	Bit 60: FIFO half full, Bit 61: FIFO full
[62]	Downstream Marker	0: Upstream Packet, 1: Downstream Packet

[63]	Parity Bit	0 if number of 1s in bits [62:0] is odd (odd parity)

Table 16. Structure of LArPix-v2b output word (data mode).

If the **enable_fifo_diagnostics** bit in the Configuration Register Map is set, then bits [43:32] will contain the number of events sitting in the shared FIFO of the chip from which the event is received at the time the event was registered. Bits [45:44] will contain number of events sitting in the given channel's local FIFO at the time the event was registered, and bits [47:46] will be 00.

If the **mark_first_packet** bit is set, then the MSB of the timestamp with be set to '1' for the first packet, and '0' for other packets in a burst.

17.2 Configuration Packet

LArPix-v2b is programmed by setting the Chip ID, Register Map Address, and Register Map Data to the desired values, setting the Packet Declaration to 10 and then sending the word to LArPix-v2b. Setting the Chip ID and Register Map Address to the desired values and then setting the Packet Declaration to 11 will return the value of the Register Map Data for the chip and address identified. The LArPix-v2b event output format (configuration mode) is shown in Table 17. Note that bit[1] = 1 indicates a configuration packet, and given a configuration packet, bit[0] = 0 indicates write and bit[0] = 1 indicates read.

Bit Range	**Contents**	**Comment**
[1:0]	Packet declaration	00: data 01: test **10: configuration write** **11: configuration read**
[9:2]	Chip ID	Each chip in a Hydra Network needs a unique ID
[17:10]	Register Map Address	
[25:18]	Register Map Data	

[61:26]	All zeros	
[62]	Downstream Marker	0: Upstream Packet, 1: Downstream Packet
[63]	Parity Bit	0 if number of 1s in bits [62:0] is odd (odd parity)

Table 17. Structure of LArPix-v2b output word (configuration mode).